AGRIBANK ONLINE

AI-Powered Fraud Detection and Risk Scoring System for Agribank Online Transactions

Technical Capstone Report

Project Domain	Finance / Financial Fraud Detection
Machine Learning Task	Supervised Binary Classification with Risk Scoring
Dataset	PaySim Synthetic Financial Dataset (Kaggle)
Target Variable	isFraud
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Program / Course	Artificial Intelligence and Machine Learning
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ACADEMIC CAPSTONE: PROOF OF CONCEPT USING SYNTHETIC DATA

This document contains no Agribank customer or transaction data.

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PART I: EXECUTIVE SUMMARY

A. Background

Agribank is preparing for its transition to a commercial bank, and digital banking is a large part of that plan. One of the initiatives for 2027 is the addition of fund transfers to Agribank Online, covering both real-time and batch transfers through InstaPay and PESONet.

These services will make banking more convenient for customers. They will also expose the Bank to more fraud and unauthorized transfers. As transaction volumes grow, manual review alone will not be enough to identify suspicious activity in time.

The Bangko Sentral ng Pilipinas (BSP) expects banks to have automated, real-time fraud monitoring. For covered institutions and products, BSP also requires a Fraud Management System (FMS) that can detect, prevent and block suspicious transactions. The controls it cites include transaction velocity monitoring, changes to device and account information, geolocation, blacklist screening and detection of unusual behavior.

B. Proposed Solution

This project proposes a fraud detection and risk scoring model as one component of a future Agribank FMS. It does not attempt to build the full FMS.

For each transaction, the model estimates the probability of fraud and converts it into a risk score from 0 to 100. Each score comes with the main factors that produced it. Fraud personnel can then decide which transactions to review first. The final decision stays with them.

C. Approach

The project uses PaySim, a public dataset of about 6.36 million simulated mobile-money transactions. It contains no Agribank data. Its purpose is to test the method, not to estimate how a model would perform on the Bank's actual transactions.

Fraud makes up only about 0.13% of the records. At that rate, a model that labels every transaction as legitimate would be 99.87% accurate and still catch no fraud. Accuracy is therefore not the main measure. The models are judged on how much fraud they catch, how many of their alerts are false, and whether the alert volume is something a review team could handle.

PaySim also has a known weakness. A few patterns created by the simulator identify fraud almost perfectly, and these are unlikely to appear in real bank data. Results are therefore reported with and without these patterns, so that the findings are not overstated.

D. Expected Outputs

- A documented, repeatable workflow published in a GitHub repository.
- Results for four machine learning models, compared against a simple rule-based benchmark.
- A 0 to 100 risk score, with an analysis of possible alert thresholds.
- The main factors behind each high-risk score.
- A review of which data fields Agribank would actually have when a transaction is scored.
- A check of whether model errors fall more heavily on certain groups of transactions.
- A list of limitations and recommended next steps.

E. Key Results

- **Headline result.** With all PaySim features, all four models reached a test PR-AUC close to 1.000 (XGBoost: 1.0000). This mostly reflects how the simulator creates fraud.
- **Realistic result.** With only the details Agribank would have when a transfer is submitted, the best model, XGBoost, reached a test PR-AUC of 0.5777 (95% interval: 0.562 to 0.594). This is about 45 times the rule-based benchmark (0.0129).
- **Review workload.** At an assumed capacity of 150 reviews a day, the realistic model caught 44.8% of fraud cases and 77.9% of fraud value, and 75.0% of its alerts were fraud.

- **Fairness.** The largest false alert disparity was for Night (time of day), at 37.4 times the overall false positive rate. A mitigation was tested and is reported in PART XI.

PART II: INTRODUCTION AND BUSINESS CONTEXT

A. Agribank's Digital Banking Direction

Agribank's move toward commercial banking will require a wider range of digital services. Fund transfers through Agribank Online are planned for 2027, using InstaPay for real-time transfers and PESONet for batch transfers. Both are part of the National Retail Payment System (NRPS), which BSP established to promote safe, efficient and reliable retail payments.

Wider digital access also creates more openings for fraud, such as stolen credentials, account takeover, social engineering and unusual transaction patterns. Reviewing transactions only after they are completed will not be enough. The Bank will need a way to flag suspicious activity while transactions are being processed and to route it quickly to the personnel who handle fraud investigation.

B. Regulatory Context

BSP Circular No. 1213. Banks are expected to protect customers from fraud committed through electronic channels, including through automated and real-time fraud monitoring. For covered institutions and products, the circular calls for an FMS that can quickly detect, prevent and block suspicious or fraudulent transactions.

BSP Memorandum No. M-2024-029. BSP-supervised institutions are reminded to maintain adequate risk management systems and controls, including an FMS, to protect customer accounts under the Anti-Financial Account Scamming Act (AFASA), Republic Act No. 12010.

These requirements are the main business reason for exploring machine learning as part of the Bank's fraud controls.

C. Project Scope

In Scope	Out of Scope
Fraud classification using the PaySim dataset	Use of actual Agribank customer or transaction data
Feature creation, feature selection and PCA testing	Device, IP address, geolocation and login analytics
Comparison of four models and a rule-based benchmark	Integration with Agribank Online, InstaPay or PESONet
Risk scoring and alert threshold analysis	Automatic blocking or declining of transactions
Model explanations and a review of error rates by group	Case management and customer notification
A sample alert list for manual review	Production approval and regulatory validation

PART III: PROBLEM UNDERSTANDING AND FRAMING

A. Business Problem

Once fund transfers go live, the number and variety of transactions to monitor will increase. Manual review will still be needed, but it becomes harder to rely on as the main control as volumes grow.

Problem statement. *"How can machine learning help identify potentially fraudulent Agribank Online transactions, and assign a risk score that helps fraud personnel decide which transactions to review first?"*

The model is meant to support fraud personnel and the Bank's existing controls. It will not make final fraud determinations or block transactions on its own.

B. Data Science Problem

Problem type. The task is supervised binary classification. The model learns from past transactions already labeled as legitimate (0) or fraudulent (1) in the isFraud column, then predicts the label for new transactions.

Scoring point. Each transaction is scored individually, before it is completed. The model can therefore use only information the Bank would have at that moment. PART VI applies this rule to every feature.

C. Research Questions

No.	Research Question
RQ1	Which transaction and balance characteristics best separate fraudulent from legitimate transactions?
RQ2	How much better does machine learning perform than the simple rule already in the dataset (isFlaggedFraud)?
RQ3	How much do the results depend on balance patterns that may be simulator artifacts, or that Agribank could not see for transfers to other banks?
RQ4	Which alert threshold gives a workable balance between fraud caught and review workload?
RQ5	Do false alerts and missed fraud occur at noticeably different rates across groups of transactions?

D. Project Objectives

1. Primary Objective

To develop and evaluate a machine learning prototype that identifies and ranks potentially fraudulent online banking transactions for manual review.

2. Specific Objectives

- Analyze a large simulated transaction dataset and identify the characteristics of fraudulent transactions.
- Clean, prepare and explore the data using a repeatable workflow.
- Build features from transaction and balance data, and confirm that each would be available at the time of scoring.
- Develop several classification models and compare them with a rule-based benchmark.
- Address the low share of fraud in the data without letting test data influence training.
- Evaluate the models using measures suited to fraud detection rather than accuracy.
- Measure how much the results depend on patterns that may not hold in real bank data.
- Explain the factors behind the model's predictions, overall and for individual transactions.
- Check whether errors are distributed unevenly across groups of transactions.
- Develop a risk score and threshold approach that supports manual review.
- Show how the model could fit into a future Agribank FMS.

PART IV: SUCCESS METRICS

A. Technical Metrics

Because fraud is so rare in the data, a model that labels everything as legitimate would still score very high on accuracy. Accuracy is reported for completeness but is not used to compare models. The following measures are used instead.

Metric	What It Measures	Why It Matters
Precision	Of the transactions flagged, the share that are actually fraud	Low precision means many false alerts and wasted review time
Recall	Of all fraud, the share the model flags	Missed fraud means financial loss and customer harm
F1-score	A single figure that combines precision and recall	Useful when both types of error matter
PR-AUC (main metric)	Overall precision and recall across all possible thresholds	Gives a more accurate picture than ROC-AUC when fraud is rare
ROC-AUC (supporting)	How well the model ranks fraud above legitimate transactions	Widely reported in other studies, but tends to look favorable when fraud is rare
False Positive Rate	Share of legitimate transactions wrongly flagged	Indicates the effect on customers and on review workload
Precision@k / Recall@k	Results among the k highest-risk transactions in a period	Reflects a review team that can only handle a fixed number of alerts
Brier score and calibration chart	Whether predicted probabilities match actual fraud rates	Needed for the 0 to 100 score to mean what it says
Confusion matrix	Counts of correct and incorrect predictions by type	Shows the actual outcomes behind the other metrics

B. Business KPIs

KPI	Definition
Fraud Detection Rate	Share of fraudulent transactions correctly flagged
Fraud Value Detected	Total value of fraudulent transactions correctly flagged
Alert Volume	Number of transactions referred for review at a given threshold
False Alert Volume	Number of legitimate transactions referred for review
Investigation Efficiency	Share of reviewed alerts confirmed as fraud
Potential Fraud Loss Exposure	Total value of fraudulent transactions the model did not flag

C. Cost of Alerts and Missed Fraud

Every threshold involves a trade-off. Missed fraud results in losses, while every alert takes up review time. Both are combined in a simple cost measure:

Expected Cost = value of missed fraud + (number of alerts × review cost per alert)

The actual review cost per alert at Agribank is not known. The analysis therefore uses a range of assumed costs and shows how the preferred threshold changes.

D. Basis for Selecting the Final Model

PR-AUC is the main comparison measure. To keep the test set clean, the model, the calibration method and the cost-based threshold are all chosen using the validation set. The test set is used once, to report results. The final choice does not rest on one figure. Recall within a fixed alert limit, the reliability of the risk scores, consistency across the stress tests, ease of explanation and processing time are also considered.

PART V: DATA COLLECTION AND UNDERSTANDING

A. Dataset Selection

The project uses the Synthetic Financial Datasets for Fraud Detection (PaySim) dataset from Kaggle, file PS_20174392719_1491204439457_log. It was selected for the following reasons:

- **Relevance.** It was built specifically for research on financial transaction fraud.
- **Size.** It has enough records to support meaningful model development and testing.
- **Available fields.** Its amount and balance fields can be used to build features that describe transaction behavior.
- **Data privacy.** Because the data is simulated, no Agribank customer information is processed.

Note: The dataset simulates mobile-money transactions. It does not represent Agribank's customers, fraud rates, payment channels or operating environment. Results should not be read as expected performance at the Bank.

B. Dataset Overview

The dataset contains 6,362,620 transactions and 11 columns. Each time step represents one simulated hour, and the data covers 743 steps, or about 30 days. There are five transaction types: PAYMENT, TRANSFER, CASH_IN, CASH_OUT and DEBIT. Amounts are in an unspecified simulated currency.

The five money fields are summarized below. All amounts and balances are in simulated currency units.

Column	Median	Mean	Skewness	Share of Zeros
amount	74,871.94	179,861.90	31.0	0.0%
oldbalanceOrg	14,208.00	833,883.10	5.2	33.0%
newbalanceOrig	0.00	855,113.67	5.2	56.7%
oldbalanceDest	132,705.66	1,100,701.67	19.9	42.5%
newbalanceDest	214,661.44	1,224,996.40	19.4	38.3%

In every field the mean is above the median and the skewness is positive. The values are heavily right-skewed, with a long tail of very large amounts. This is why amounts are log-transformed in PART VI and why extreme values are kept rather than removed (see Data Quality Checks below).

C. Data Dictionary

The table lists each column with its type, range and unit, role, and whether it would be available to Agribank at the time of scoring. "On-us" refers to a transfer between two Agribank accounts. "Interbank" refers to a transfer to another bank through InstaPay or PESONet.

Column	Type	Description	Range and Unit	Role	Available to Agribank at Scoring
step	Integer	Simulated hour	1 to 743 (hours)	Input	Yes (transaction timestamp)
type	Category	Transaction type	PAYMENT, TRANSFER, CASH_OUT, CASH_IN, DEBIT	Input	Yes
amount	Numeric	Transaction amount	0.00 to 92,445,516.64 (simulated currency)	Input	Yes
nameOrig	Identifier	Sender account	Text ID starting with "C"	Identifier	Yes, but used only for history features

Column	Type	Description	Range and Unit	Role	Available to Agribank at Scoring
oldbalanceOrg	Numeric	Sender balance before the transaction	0.00 to 59,585,040.37 (simulated currency)	Input	Yes, since Agribank holds the account
newbalanceOrig	Numeric	Sender balance after the transaction	0.00 to 49,585,040.37 (simulated currency)	Input (derived)	Yes, as the old balance less the amount
nameDest	Identifier	Receiving account	Text ID starting with "C" (customer) or "M" (merchant)	Identifier	Yes (used for receiving account history)
oldbalanceDest	Numeric	Receiving balance before the transaction	0.00 to 356,015,889.35 (simulated currency)	Input	On-us transfers only
newbalanceDest	Numeric	Receiving balance after the transaction	0.00 to 356,179,278.92 (simulated currency)	Input	No. Known only after the transfer, and never for other banks.
isFraud	Binary	Fraud label	0 = legitimate, 1 = fraud	Target	No. Known only after investigation.
isFlaggedFraud	Binary	Output of the simulator's fraud rule	0 or 1	Benchmark only	Not used as an input

D. Known Characteristics of the Dataset

Exploration confirmed the patterns reported in earlier analyses of PaySim. Several of them make fraud much easier to detect than it would be in real bank data.

- **Fraud by type.** All 8,213 fraud cases are TRANSFER or CASH_OUT transactions. No fraud appears in the other three types.
- **Emptied accounts.** 97.8% of fraud cases move the full sender balance, leaving it at zero.
- **Receiving balances.** 49.6% of fraud cases show a receiving balance of zero before and after a non-zero transfer, compared with 36.4% of legitimate transactions.
- **Merchant accounts.** 33.8% of all transactions go to receiving accounts that begin with "M". These merchant accounts have no balance information.
- **Sender accounts.** 99.85% of sender account IDs are unique, so sender transaction history cannot be built.
- **Simulator rule.** isFlaggedFraud flagged only 16 transactions in the whole dataset, all of them TRANSFERs of at least 353,874. It catches almost no fraud.

E. Data Quality Checks

All checks were scripted and can be rerun from the GitHub repository. The results were as follows:

Check	Result
Records and columns	6,362,620 transactions and 11 columns
Missing values	None in any column
Duplicate records	None
Category values	Five transaction types, no unexpected values
Time coverage	743 hourly steps, about 30 days
Extreme amounts	338,078 transactions (5.3%) lie above the IQR upper bound of 501,719.34. Their fraud rate is 1.14%, against 0.129% overall. They were kept, since removing them would remove a large share of fraud.

Check	Result
Amount and balance consistency	Measured through the balance reconciliation features in PART VI
Leakage	isFlaggedFraud and post-transaction balances identified as risks (PART XII)

F. Class Imbalance

The dataset contains 8,213 fraudulent transactions, 0.129% of all records, or about 1 in every 775. The project addressed this as follows:

- Split the data by time so that each portion kept enough fraud cases (PART VIII).
- Gave fraud cases more weight during training.
- Sampled down legitimate transactions in the training set only. Validation and test data were not rebalanced.
- Evaluated models on fraud-focused measures instead of accuracy.
- Tested a range of alert thresholds instead of a fixed 50% cut-off.

PART VI: DATA PREP, EXPLORATION & FEATURE ENGINEERING

A. Data Preparation

Data preparation was done in Python and is stored in the project's GitHub repository. It covered the following steps:

- Validated data types, the fraud label and value ranges.
- Checked for missing values, duplicates and invalid entries. None were found.
- **Scope filter.** Fraud occurs only in TRANSFER and CASH_OUT, so modeling was limited to these two types.
  a. This reduced the data to 2,770,409 records.
  b. The filter is documented as a business rule, since in practice it would run before the model.
- Converted transaction types into numeric form.
- Reviewed extreme values and kept them.
- Assessed every candidate feature for leakage.
- Split the data by time into training, validation and test sets (PART VIII).

Note: Leakage happens when a model is trained using information it would not have in actual use, which makes results look better than they are. To prevent this, all data adjustments (such as scaling or rebalancing) are learned from the training set only and then applied unchanged to the validation and test sets.

B. Exploratory Data Analysis

Exploration compared fraudulent and legitimate transactions and tested the characteristics listed in PART V. It covered:

- The share of fraud overall and by transaction type.
- Transaction amounts for fraudulent and legitimate transactions.
- Sender and receiving balance behavior, including zero balances.
- Fraud volume and rate by hour and by day.
- Relationships between the columns.
- Cases where amounts and balance changes do not reconcile.

Fraud pairs. 99.5% of fraudulent TRANSFERs are matched by a fraudulent CASH_OUT of the same amount in the same hour. This matches how PaySim models fraud, where funds are moved to another account and then cashed out.

Amounts. Within TRANSFER and CASH_OUT, the median fraudulent amount is 441,423.44, against 171,034.46 for legitimate transactions (Figure 1). Fraud amounts also show a sharp spike at exactly 10,000,000, the largest fraudulent amount in the data. This looks like another simulator trait rather than real fraud behavior.

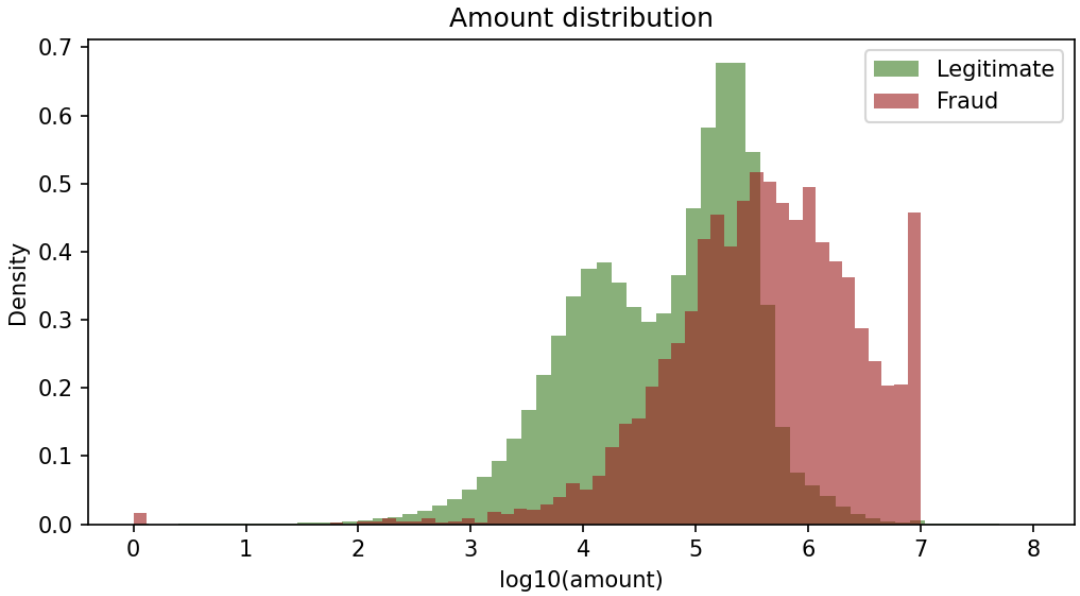


Figure 1. Transaction amounts, fraudulent and legitimate (log scale)

Time of day. Fraud makes up 1.77% of transactions between 00:00 and 05:59, against 0.10% for the rest of the day. Legitimate activity is concentrated in daytime hours while fraud is spread more evenly across the day (Figure 2). This pattern returns in the bias review in PART XI.

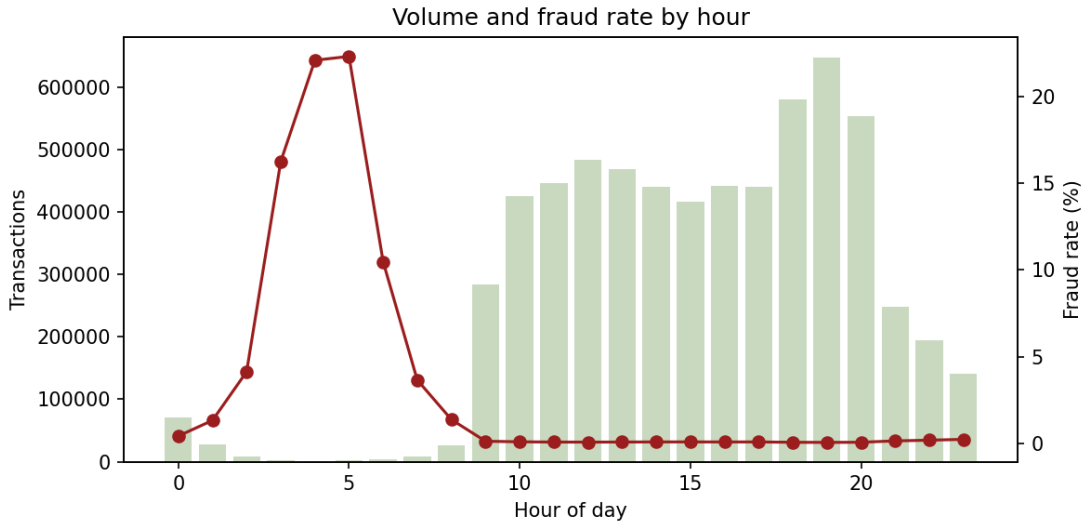


Figure 2. Transaction volume and fraud rate by hour of day

Volume over time. Legitimate transactions averaged 347,112 a day in the first half of the month and 53,374 in the second half. Fraud averaged 266 and 263 a day over the same periods (Figure 3). This is why the later test period has a higher fraud rate than training (PART VIII).

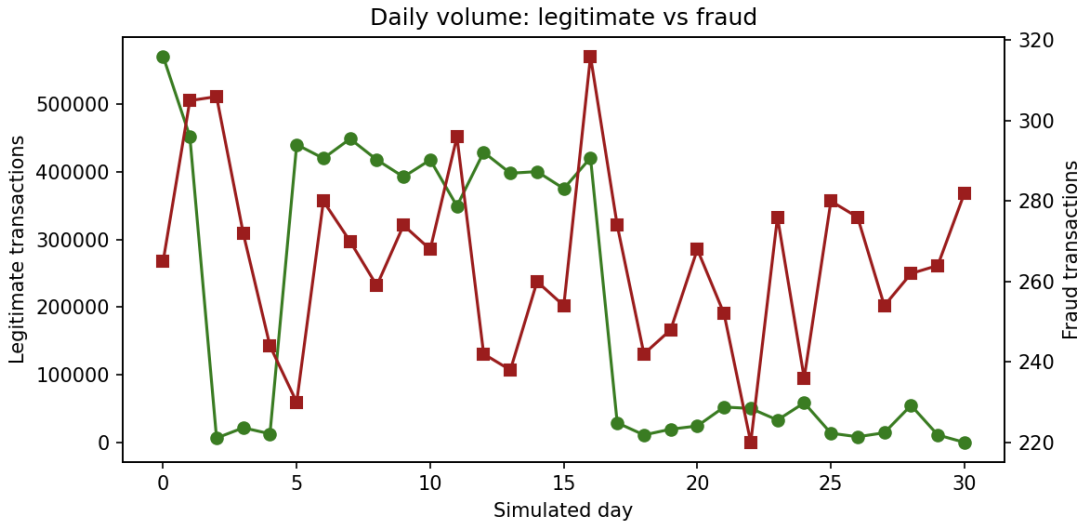


Figure 3. Daily legitimate and fraud volume

Correlations. Sender balances before and after a transaction are highly correlated, as are receiving balances before and after. No single raw field is strongly correlated with the fraud label (Figure 4). Fraud can therefore not be found from one field, which is why derived features are needed.

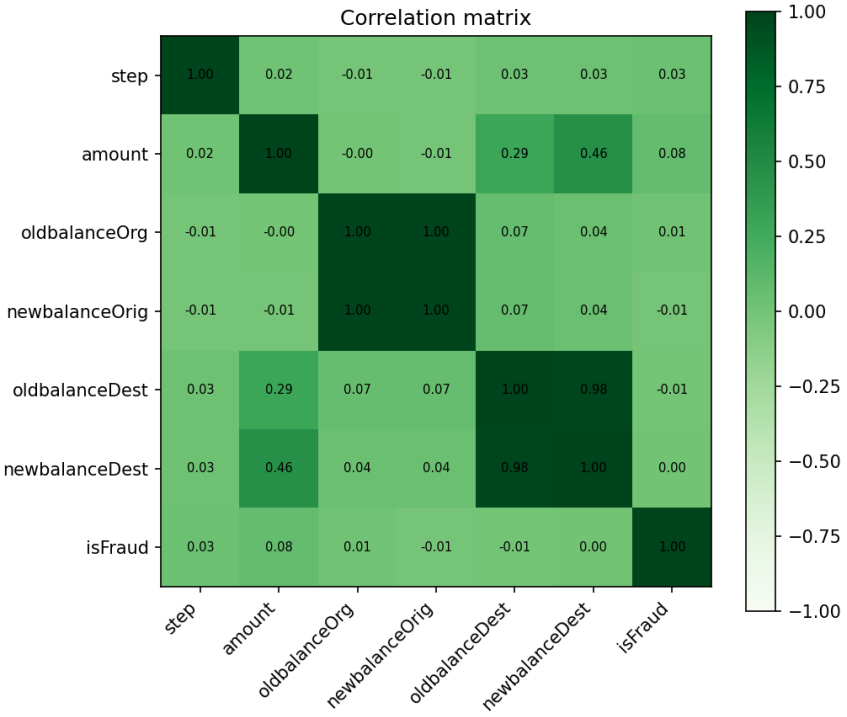


Figure 4. Correlation matrix of numeric fields

All charts and tables are saved in the reports folder of the repository.

C. Feature Engineering

Features are additional fields derived from the raw data. The aim is to describe transaction behavior, not just restate the original values.

1. Balance Reconciliation Features

After a transfer, each balance should change by exactly the transaction amount. Two features measure any difference:

Sender difference = new sender balance + amount − old sender balance

Receiver difference = old receiving balance + amount − new receiving balance

Two indicators support these: one for a sender account emptied by the transaction, and one for a receiving balance that remains zero after a non-zero transfer.

Note: Earlier studies found that these features alone identify nearly all PaySim fraud. This is likely a result of how the simulator works. Their effect is measured separately in the stress tests in PART VIII.

2. Amount Features

- Transaction amount on a logarithmic scale, to reduce the effect of very large values.
- Amount as a share of the sender balance.
- Amount as a share of the receiving balance (on-us transfers only).
- Indicator for high-value transactions.
- Difference between the amount and the typical amount for its transaction type.

3. Time Features

- **Hour of day.** Derived from the step number (remainder after dividing by 24).
- **Day.** The step number divided by 24, rounded down.
- **Period of day.** Night, morning, afternoon or evening.

The dataset covers only about one month, so weekly or seasonal features are not used.

4. Transaction Type Features

After the scope filter only two types remain, so transaction type is represented by one indicator field (1 for TRANSFER, 0 for CASH_OUT). Its contribution is measured through the SHAP analysis in PART X.

5. Receiving Account History

Sender accounts almost never repeat in PaySim, so sender history features are not possible. Receiving accounts do repeat. Three features are built from their history: the number of earlier transfers the account received, the total amount it received (log scale), and an indicator for an account with no earlier transfers. Only transactions in earlier hours are counted, so no future information is used.

D. Feature Availability Review

A feature is only usable if the Bank would have it when the transaction is scored. Each feature group is reviewed below.

Feature Group	Source	On-us Transfer	Interbank Transfer	Treatment
Amount, type and time	Transfer request	Available	Available	Used in both sets
Sender balance features	Agribank account records	Available	Available	Used in both sets
Receiving balance features	Receiving account records	Available	Not available	Feature Set A only
Receiving account history	Earlier transfers to the account	Available	Partial (Agribank records only)	Tested in both sets
isFlaggedFraud	Simulator rule	Not applicable	Not applicable	Benchmark only

Five feature sets were built from this review, each used in one or more of the stress tests in PART VIII. Sets A and C include the receiving balance after the transfer, which is not known at scoring time. Set D is the realistic view, and it is also used for the model comparison in PART VIII, Section G.

Feature Set	Contents	Used In
A (full PaySim)	All 21 features, including the receiving balance after the transfer	E1, E4
A without reconciliation	Set A less the four balance reconciliation features (17)	E2
B (interbank)	Set A less the receiving balance features (16)	E3
C (no sender balances)	Set A less reconciliation and sender balance fields (14)	E5
D (transaction details only)	Amount, type, time and receiving account history only (11)	E6

PART VII: FEATURE SELECTION AND DIMENSIONALITY REDUCTION

A. Feature Selection

Two methods were used to identify which features are worth keeping:

- **Mutual information.** Measures how much each feature says about fraud, independent of any model. It can detect relationships that are not linear.
- **Model-based importance.** Ranks features using Random Forest or XGBoost. The ranking is checked by shuffling one feature at a time on the validation set and measuring the drop in performance (permutation importance).

A feature would be removed if it added little information, duplicated another feature, carried a leakage risk, or would not be available at the time of scoring.

Results. Features weak on all three measures, and therefore removed: None. The top-ranked feature was orig_emptied by mutual information, orig_emptied by Random Forest importance and orig_emptied by permutation importance. Permutation importance was close to zero for most features. This does not mean those features are useless. The model was already near perfect on Feature Set A, so other features made up for any single feature that was shuffled.

Correlated features. Two pairs were highly correlated: the receiving account history fields log_dest_prior_amount and dest_is_new (0.97), and the two receiving balance fields (0.96). Both pairs were kept, since tree-based models are not harmed by correlated inputs. Removing one field from each pair is an option for a later iteration.

B. Dimensionality Reduction

Principal Component Analysis (PCA) combines related numeric features into a smaller number of components. It was fitted on the training data after scaling. The analysis reports how much variation the components capture, plots fraudulent and legitimate transactions in two dimensions, and tests whether the components improve Logistic Regression.

PCA components are difficult to explain to fraud personnel, so they were to be used only if they clearly improved results. Since the feature set is small, the improvement was expected to be limited, and tree-based models were not expected to benefit.

Results. 14 of 21 components were needed to capture 95% of the variation. Logistic Regression with default settings reached a validation PR-AUC of 0.9999 without PCA and 0.9891 with it. PCA was therefore not adopted, and the original features were kept. The tuned Logistic Regression in PART VIII uses different settings, so its score differs slightly.

PART VIII: MODEL DEVELOPMENT AND EVALUATION

A. Testing Approach

In actual use, a model is trained on past transactions and then scores new ones. The data was split by time to reflect this:

Data Set	Time Steps	Records	Fraud Cases	Fraud Rate
Training	1 to 322	1,938,484	3,633	0.19%
Validation	323 to 376	415,628	560	0.13%
Test	377 to 743	416,297	4,020	0.97%

The split follows the first 70%, next 15% and last 15% of records in time order. Legitimate activity falls off sharply in the later part of the simulated month while fraud continues (Figure 3). As a result, the test period covers about half of the month, and its fraud rate is about five times that of training and about seven times that of validation. This affects calibration (PART IX) and is noted as a limitation in PART XII.

A random split was also run for comparison (test E4).

B. Handling Class Imbalance

- Class weighting gave fraud cases more weight during training.
- Legitimate transactions in the training set were sampled down to 10%, leaving 197,118 records with a fraud share of 1.84%.
- SMOTE was available but not used in the final run, because class weighting and sampling were sufficient.
- The validation and test sets were not rebalanced.

C. Candidate Models

Model	Role	Reason for Inclusion
Rule-based benchmark (isFlaggedFraud)	Benchmark	Shows how a simple fixed rule performs. Answers RQ2.
Logistic Regression	Baseline model	Simple and transparent, with results that are easy to interpret
Decision Tree	Rule-based model	Produces decision rules that fraud personnel can follow
Random Forest	Ensemble model	Captures more complex patterns and provides feature importance
XGBoost	Gradient boosting model	Performs well on structured data and handles class imbalance directly

D. Model Tuning

On Feature Set A, each model was tuned with a random search of up to eight settings, scored on the validation set using PR-AUC. Set A is close to saturated, so more tuning there would not change the ranking. Tuning effort was therefore focused on Feature Set D (Section G). Settings, random seeds and software versions are saved in the repository (ANNEX E). The selected settings on Set A were:

Model	Selected Settings	Validation PR-AUC	Tuning Time
Logistic Regression	C 0.01	1.0000	1 seconds
Decision Tree	min samples per leaf 1, max depth 4	0.9982	10 seconds

Model	Selected Settings	Validation PR-AUC	Tuning Time
Random Forest	trees 400, min samples per leaf 1, features per split sqrt, max depth no limit	1.0000	115 seconds
XGBoost	row sample 1.0, trees 600, min child weight 1, max depth 8, learning rate 0.05, column sample 0.7	1.0000	12 seconds

E. Comparison of Results

All models were evaluated once on the same test set, using Feature Set A. The table shows results at a threshold of 0.5.

Model	PR-AUC	ROC-AUC	Precision	Recall	Alerts
Rule-based benchmark	0.0129	0.5016	1.0000	0.0032	13
Logistic Regression	0.9985	0.9990	1.0000	0.9980	4,012
Decision Tree	0.9995	0.9996	0.9875	0.9995	4,069
Random Forest	1.0000	1.0000	1.0000	0.9998	4,019
XGBoost	1.0000	1.0000	0.9995	0.9998	4,021

All four models reached near-perfect scores, and the differences between them are too small to matter. The rule-based benchmark caught 13 of the 4,020 fraud cases in the test period, worth about 71 million of 6.33 billion in fraud value (simulated units). This answers RQ2: on this dataset, machine learning performs far better than the simple rule.

Overfitting. Training and validation PR-AUC differed by no more than 0.008 for any model, so there is no sign of overfitting. Training scores were measured on the sampled training set, which is why some are slightly below the validation scores.

F. Stress Tests

These tests address RQ3. They show how much performance depends on features that may not be reliable or available in practice. Each test used the selected model (XGBoost) with the same settings.

Test	Setup	Features	PR-AUC	Recall	Precision
E1	Feature Set A, time-based split	21	1.0000	0.9998	0.9995
E2	Set A without reconciliation features	17	0.9994	0.9990	0.9812
E3	Feature Set B (interbank)	16	0.9998	0.9998	0.9963
E4	Feature Set A, random split	21	0.9967	0.9935	0.9895
E5	Feature Set C (no sender balances)	14	0.7945	0.7968	0.4633
E6	Feature Set D, Set A settings	11	0.5568	0.7808	0.1862
E6 tuned	Feature Set D, tuned on Set D (XGBoost)	11	0.5777	0.8423	0.1334

Note: Recall and precision are at a threshold of 0.5, before calibration. E1 to E6 all use the XGBoost settings tuned on Set A, so only the features change between tests. The last row uses the model tuned on Set D (Section G).

The model rebuilt the simulator signal. Removing the four reconciliation features (E2) had almost no effect. The model found the same "account emptied" signal in the sender balance fields that remained. With sender balances removed (E5), it switched to the receiving balance after the transfer, a field that would not be known at scoring time.

E6 is the realistic result. With only the details Agribank would have when a customer submits a transfer, PR-AUC fell to 0.557 with the Set A settings. After tuning on Set D, the selected model reached 0.578. That is far below the headline result, but about 45 times the rule-based benchmark (0.0129).

Interbank transfers. Removing receiving balances (E3) made little difference. The model's dependence was on the sender side, not on the receiving balances that are unavailable for interbank transfers.

Split method. The random split (E4) scored slightly lower than the time-based split (E1). The two test sets have different fraud rates, so this comparison is less informative than planned.

G. Model Comparison on the Realistic Feature Set

On Feature Set A all four models scored close to 1.000, so that comparison cannot separate them. The four models were therefore tuned again on Feature Set D, the realistic view, with a random search of up to 20 settings each. Selection used validation PR-AUC only. The test set was used once, to report the results below.

Model	Selected Settings (Set D)	Validation PR-AUC	Tuning Time
Logistic Regression	C 10	0.0792	1 seconds
Decision Tree	min samples per leaf 20, max depth 6	0.3421	12 seconds
Random Forest	trees 200, min samples per leaf 5, features per split 0.5, max depth 8	0.3172	299 seconds
XGBoost	row sample 1.0, trees 400, min child weight 5, max depth 3, learning rate 0.05, column sample 0.7	0.3696	27 seconds

Test results are shown with a 95% bootstrap interval for PR-AUC, based on 200 resamples of the test set. Recall@150 and Precision@150 use the capacity rule: the 150 highest scores in each day are referred for review.

Model	PR-AUC	95% Interval	ROC-AUC	Recall@150	Precision@150
Rule-based benchmark	0.0129	Not applicable	0.5016	Not applicable	Not applicable
Logistic Regression	0.2694	0.256 to 0.284	0.8688	31.3%	52.5%
Decision Tree	0.5260	0.509 to 0.541	0.9328	43.4%	72.7%
Random Forest	0.5562	0.539 to 0.570	0.9416	43.9%	73.5%
XGBoost	0.5777	0.562 to 0.594	0.9422	45.3%	75.9%

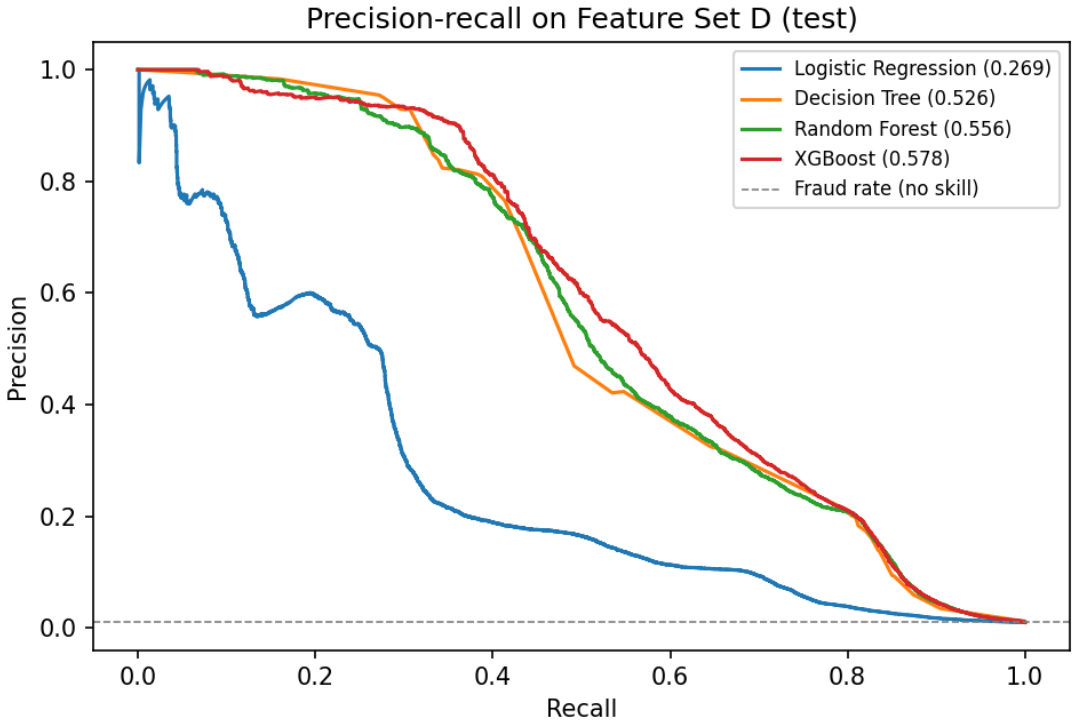


Figure 5. Precision-recall curves on Feature Set D (test period)

Selection. XGBoost had the highest validation PR-AUC, and no other model came within 0.002 of it, so it was selected on that basis. Where the 95% intervals of two models overlap, the difference between them should not be treated as meaningful.

Overfitting. The largest gap between training and validation PR-AUC on Set D was 0.362 (Random Forest). Training scores were measured on the sampled training set, where fraud makes up about 1.8% of records, against 0.13% in validation. PR-AUC rises with the fraud share, so much of this gap comes from the different fraud share rather than from memorization. Random Forest was not selected in any case.

H. Selecting the Final Model

Headline model (Feature Set A). All four models scored within 0.0018 of each other on validation PR-AUC, so the choice was made on the other criteria in PART IV. XGBoost was selected for the following reasons:

- Its scores were tied with the best models on both validation and test data.
- SHAP can explain its predictions directly, so the risk score and the reasons shown to fraud personnel come from the same model.
- It tuned in 12 seconds, compared with 115 seconds for Random Forest.
- It handles class imbalance directly through its weighting setting.

Realistic model (Feature Set D). XGBoost was selected on validation PR-AUC (Section G). PARTs IX to XI use this model for the realistic view, labeled E6. The headline XGBoost model is used for E1, and its settings for E2 and E5.

PART IX: RISK SCORING AND THRESHOLD ANALYSIS

A. Score Calibration

Class weighting and rebalancing distort a model's raw probabilities. A raw output of 0.7 may not mean a 70% chance of fraud. Calibration corrects this using the validation set. Platt scaling and isotonic regression were compared, and results were checked with a calibration chart and the Brier score.

Results. The calibration method was chosen by a 5-fold Brier score inside the validation set, so the test set played no part in the choice. Isotonic regression was selected for all four views (E1, E2, E5 and E6).

Calibration when fraud rates change. Calibration was fitted on the validation period, where fraud was 0.13% of transactions. The test period had 0.97%. As a result, the E6 scores understated risk in the test period. The average E6 score was 0.42%, and the Low risk level still contained 2,505 fraud cases (Figure 6). A score calibrated in one period can be wrong in another when the fraud rate shifts. For this reason, the capacity rule is used as the main operating rule (Section D), and calibration would need regular review in production.

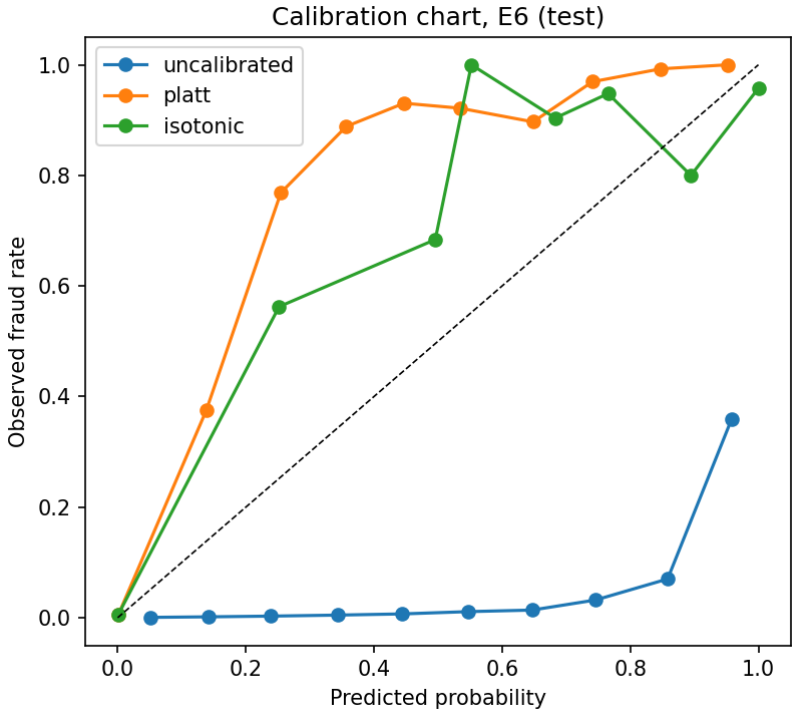


Figure 6. Calibration chart for E6 (test period)

Note: Calibration can shift the ranking measures slightly. Isotonic regression groups scores into steps, which creates ties among transactions. For E6, PR-AUC is 0.5777 before calibration and 0.5359 after, and recall at 150 reviews a day is 45.3% before and 44.8% after. PART VIII reports scores before calibration. PARTs IX and XI report calibrated scores.

B. Risk Score

Risk Score = calibrated probability of fraud × 100

The result is a score from 0 to 100 that can be grouped into review levels.

C. Risk Levels

Score-based levels. Fixed ranges assign each transaction to a risk level based on its score:

Risk Level	Score Range (illustrative)	Proposed Handling
Low	0 to 29	Normal processing and monitoring
Medium	30 to 69	Additional monitoring
High	70 to 100	Priority manual review

Capacity-based levels. The k highest-scoring transactions in each day are referred for review, with k set to the number of cases the review team can handle. This keeps the workload predictable. Because it ranks transactions instead of applying a fixed cut-off, it is not affected when the fraud rate shifts.

Note: These ranges are for the prototype only. Actual thresholds would need to be based on Agribank transaction data, fraud loss information, staffing capacity, customer impact and approved risk management policies.

Results for E6. Applying the score ranges to the E6 model in the test period gave the following results:

Risk Level	Transactions	Fraud Cases	Fraud Rate
Low	414,530	2,505	0.60%
Medium	1,054	834	79.13%
High	713	681	95.51%

Capacity-based results. The test period averaged about 263 fraud cases per day over 15.3 days. The table shows E6 results at three daily review capacities. These capacities are assumptions for the prototype, not Agribank staffing figures.

Daily Review Capacity	Recall	Precision	Fraud Value Caught	Total Alerts
50	18.1%	91.0%	38.3%	800
150	44.8%	75.0%	77.9%	2,400
300	62.8%	52.8%	90.4%	4,782

If daily capacity is below the daily number of fraud cases, even a perfect model cannot catch all fraud. At 50 reviews a day, recall is capped at about 20%.

D. Threshold Analysis

Each threshold was assessed for fraud recall, precision, alert volume, false alerts, fraud value detected and expected cost. The table shows E6, the realistic view, in the test period. It is for comparison only. No threshold was chosen using the test set.

Threshold	Recall	Precision	Fraud Value Caught	Alerts per Day	False Alerts (total)
0.01	79.6%	21.4%	97.1%	About 978	11,761
0.02	64.8%	37.3%	92.5%	About 456	4,369
0.05	47.5%	64.2%	84.4%	About 195	1,066
0.10	37.9%	85.5%	71.4%	About 117	259
0.50	28.5%	93.4%	55.3%	About 80	81
0.90	14.6%	95.8%	34.6%	About 40	26

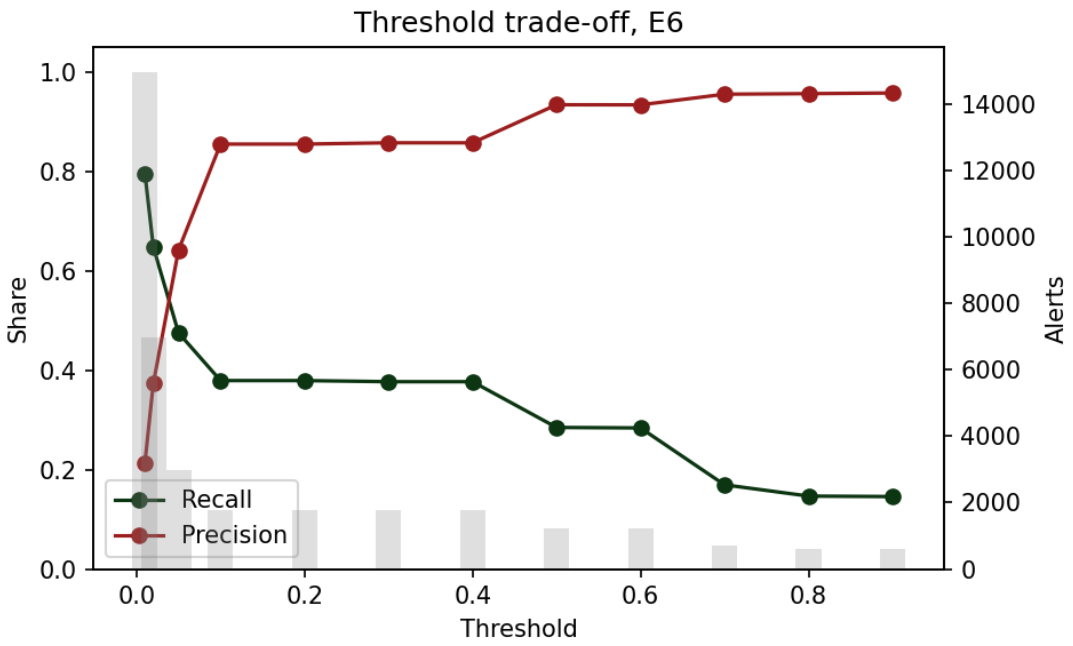


Figure 7. Recall, precision and alert volume by threshold for E6 (test period)

The cost measure favors the lowest threshold. On the validation set, expected cost was lowest at 0.01, the lowest threshold tested. The test set gives 0.01, shown for reference only. Each fraud case is worth far more than the assumed cost of a review, so the formula keeps favoring more alerts. It does not account for review capacity or for the inconvenience to customers whose transfers are flagged.

Proposed operating rule. The prototype uses the capacity rule as its main operating point: the 150 highest-scoring transactions each day are referred for review. In the test period this caught 44.8% of fraud cases and 77.9% of fraud value, and 75.0% of alerts were fraud. A fixed threshold was not used as the operating point, because its alert volume moves with the fraud rate. For comparison, a threshold of 0.05 gave about 195 alerts a day, and 0.01 gave about 978.

The choice of threshold is a business and operational decision, not only a technical one.

Note: Amounts are in simulated currency units. Review costs and capacities are assumptions for the prototype.

PART X: MODEL EXPLAINABILITY

A. Purpose

Fraud personnel need to understand why a transaction was flagged before acting on it. The explainability work answers two questions: which features the model relies on overall, and why a specific transaction received a high score.

B. Methods

- **SHAP.** Shows how much each feature raised or lowered a particular prediction. Applied to the headline XGBoost model for E1, the Set A XGBoost settings for E2 and E5, and the selected XGBoost model for E6.
- **Model coefficients.** The weight of each feature in Logistic Regression is saved for reference.
- **Permutation importance.** Used as an independent check on the validation set (PART VII).

C. Outputs

For each view, the analysis produced charts of overall feature importance, charts of how the main features affect the score, and a sample alert list. The three most influential features in each view were:

View	Top Three Features (SHAP)	What It Shows
E1	Sender account emptied; sender balance does not reconcile; amount large relative to sender balance	The model relies on the simulator's account-emptying pattern
E2	Amount large relative to sender balance; sender balance level; large amount	Without the reconciliation features, it rebuilds the same signal from sender balances
E5	Receiving balance after transfer; amount large relative to receiving balance; time of day	Without sender balances, it moves to a field not known at scoring time
E6	Time of day; amount well above usual for this type; amount previously received by account	What the model relies on when no balance fields are available

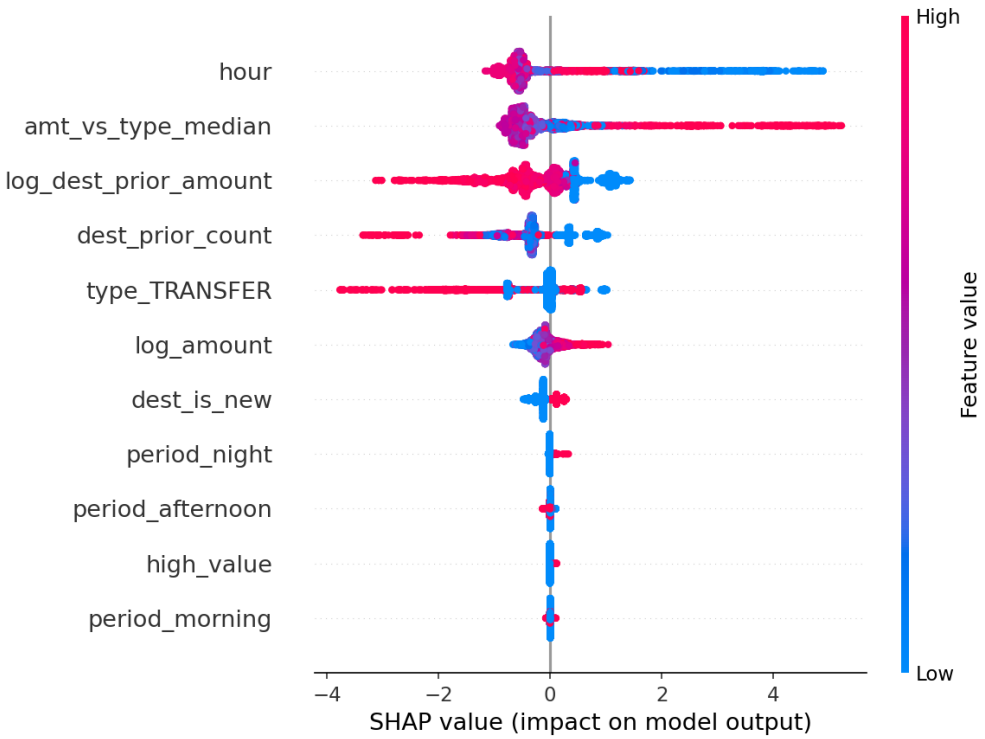


Figure 8. SHAP summary for the E6 model (each dot is one transaction)

In E6, time of day was the most important feature by mean absolute SHAP value (Figure 8). It was followed by how far the amount is above the usual amount for its type, and by the amount the receiving account had received before. Receiving account history is a plausible real-world signal, since accounts used to collect fraudulent funds often show unusual receiving patterns. Time of day is less reliable. In PaySim, fraud makes up a much larger share of night-time transactions than of daytime ones (1.77% against 0.10%), so the model learns that night-time transfers are more suspicious. The realistic model therefore leans most on a pattern that may be a simulator artifact. This is reported as a limitation in PART XII and discussed as a bias issue in PART XI.

D. Presentation to Fraud Personnel

Each high-risk alert includes up to three reasons based on the largest SHAP contributions. In the E6 sample alert list, the most common reasons were "time of day", "amount well above usual for this type" and "amount previously received by account". These are indicators for the reviewer to consider. They are not a finding that the customer committed fraud.

The main features were compared with the dataset characteristics in PART V. In E1 and E2, the model relied mostly on those patterns, which is reported as a limitation.

PART XI: ETHICAL AI AND BIAS REVIEW

A. Purpose

A false alert can delay a legitimate customer's transfer or require additional verification. Overall accuracy is not enough. The review also checks whether errors fall more heavily on particular groups.

B. Data Limitations

PaySim has no customer attributes such as gender, age, location or income. Fairness checks based on customer demographics cannot be performed, and none of the results describe Agribank's customer base.

C. Transaction Groups Reviewed

Error rates were compared across groups of transactions. For each group, the review reported the alert rate, the false positive rate (FPR), the detection rate (also called the true positive rate) and the fraud rate. The gap between the highest and lowest group shows how unevenly each kind of transaction is treated.

Group	Question Addressed
Transaction type	Does one type account for a disproportionate share of false alerts?
Amount range	Are small and large transactions treated differently?
Sender balance range	Are low-balance accounts flagged more often?
Time of day	Is activity at night flagged more often than activity during the day?
Receiving account type	Not reviewed in this run. Planned for a later iteration.

D. Results of the Bias Review

The table shows the E6 model under the main operating rule, where the 150 highest scores each day are reviewed. Each figure is the gap between the highest and lowest group. The last column compares the alert-rate gap with the fraud-rate gap.

Group	Alert Rate Gap	Fraud Rate Gap	FPR Gap	Detection Rate Gap	Alert Gap vs Fraud Gap
Transaction type	0.012	0.018	0.004	0.013	Within the fraud-rate gap
Amount range	0.014	0.019	0.002	0.351	Within the fraud-rate gap
Sender balance range	0.030	0.062	0.000	0.260	Within the fraud-rate gap
Time of day	0.177	0.155	0.054	0.628	2.1 points above the fraud-rate gap

- **Alert rates and fraud rates.** Where the alert-rate gap is within the fraud-rate gap, a group receives more alerts because it has more fraud, not because the model treats it unfairly. Where it is above, the group is flagged beyond its fraud level. The last column gives the result for each grouping.
- **Detection rates differ.** Among groups with at least 30 fraud cases, detection ranged from 19.8% for afternoon transactions to 82.6% for night-time transactions. Both extremes fall on time of day, which matches the model's reliance on that feature. Customers in groups with low detection are less protected.
- **False alerts.** Night-time transactions had the highest false positive rate of any group, at 37.4 times the overall rate. In practice, this would affect customers who bank at night, such as shift workers.
- **Transaction type.** The detection rate gap between CASH_OUT and TRANSFER fraud was 0.013.

Fairness depends on the alert rule. The review was repeated with fixed thresholds of 0.05 and 0.01. Lower thresholds raise alert volumes. The table shows whether any grouping is then flagged beyond its fraud level.

Group	Threshold	Alert Rate Gap	Fraud Rate Gap	FPR Gap	Alert Gap vs Fraud Gap
Transaction type	0.05	0.015	0.018	0.007	Within the fraud-rate gap
Transaction type	0.01	0.120	0.018	0.103	10.3 points above the fraud-rate gap
Amount range	0.05	0.016	0.019	0.003	Within the fraud-rate gap
Amount range	0.01	0.063	0.019	0.046	4.4 points above the fraud-rate gap
Sender balance range	0.05	0.032	0.062	0.001	Within the fraud-rate gap
Sender balance range	0.01	0.071	0.062	0.029	1.0 points above the fraud-rate gap
Time of day	0.05	0.200	0.155	0.081	4.4 points above the fraud-rate gap
Time of day	0.01	0.356	0.155	0.248	20.0 points above the fraud-rate gap

E. Financial Inclusion

Customers with small balances and low-value transfers are central to financial inclusion. In this review, the false positive rate gap across sender balance ranges was below 0.001, and the detection rate gap was 0.260. With real data, the Bank would need to check whether low-balance customers are less protected, more often flagged, or both.

F. Proposed Mitigations

- Limit how much weight time of day carries in the model, or remove the time features altogether. This option was tested (Section G).
- Use the capacity rule as the operating point, rather than lowering a fixed threshold to catch more fraud.
- Reweight training data for groups with low detection rates and compare the results.
- Monitor error rates by group monthly once real data is used.
- Keep a person reviewing every alert. The model does not block transactions.

G. Mitigation Test

The first mitigation was tested directly. The selected E6 model was retrained without the four time-of-day features, with all other settings unchanged. Both versions were compared under the same capacity rule of 150 reviews a day.

Measure	With Time Features	Without Time Features
Test PR-AUC	0.5359	0.4190
Recall (150 a day)	44.8%	37.8%
Precision (150 a day)	75.0%	63.2%
Alert rate gap by time of day	0.177	0.060
FPR gap by time of day	0.054	0.003
Detection rate gap by time of day	0.628	0.057

Note: The "With Time Features" column uses the calibrated E6 scores, as in PARTs IX and XI. The model without time features is not calibrated. Calibration lowers PR-AUC slightly (PART IX, Section A), so the fair comparison of PR-AUC is before calibration for both: 0.5777 with time features against 0.4190 without. The result below uses that comparison.

Result. Removing the time features narrowed the false alert gap by time of day, but PR-AUC fell by 0.159 (from 0.578 to 0.419, both before calibration). The trade-off between fairness and fraud caught is a business decision.

Reweighting groups with low detection rates was not tested in this run. It is listed as a next step.

H. Safeguards

- The model supports manual review and does not block transactions.
- Every high-risk alert comes with the reasons behind the score.
- Error rates by group are reported together with overall results.
- The limits of simulated data are stated wherever results are presented.

PART XII: DATA AND MODEL LIMITATIONS

A. Simulated Data

PaySim does not represent actual Agribank transactions. Results from this dataset cannot be used to estimate performance on Agribank Online.

B. Overstated Performance

The simulator generates fraud in consistent ways, such as emptying the sender account. The stress tests show how much the headline result depends on this:

Test	What the Model Can See	PR-AUC
E1	All PaySim features	1.0000
E2	No balance reconciliation features	0.9994
E5	No sender balance fields	0.7945
E6	Transaction details only (Set A settings)	0.5568
E6 tuned	Transaction details only (XGBoost, tuned on Set D)	0.5777
Rule	isFlaggedFraud benchmark	0.0129

Each time one source of the simulator pattern was removed, the model found another. Performance fell to a level closer to what could be expected from real data only when all balance fields were removed. Actual fraudsters change their methods, and real data is less consistent, so even E6 may overstate real performance.

C. Differences from Philippine Banking Transactions

Simulated mobile-money activity may differ considerably from InstaPay and PESONet transfers. The project demonstrates a method. It does not show readiness for production use.

D. Data Availability in Practice

For interbank transfers, Agribank cannot see the receiving account balance. Removing receiving balances (E3) had little effect, because the model relied on sender-side patterns. The receiving balance after the transfer is not known at scoring time for any transfer. It is included in Feature Sets A and C only to show its effect, and Feature Set D excludes all balance fields.

E. Small Number of Fraud Cases

Because fraud is rare, results can shift noticeably with a small number of cases. The validation set had only 560 fraud cases. The 95% bootstrap intervals in PART VIII, Section G show how much the test PR-AUC could move from sampling alone.

F. Leakage Risk

Balances recorded after the transaction and the simulator's own fraud flag can reveal the fraud label. isFlaggedFraud is used only as a benchmark, and the handling of each post-transaction balance is documented.

G. Changes in Fraud Patterns

The dataset covers about one month. Fraud patterns change over time, so a production model would need ongoing monitoring and periodic retraining. This cannot be tested with the current data.

H. Score Calibration Across Periods

Scores calibrated in the validation period understated risk in the test period, where the fraud rate was about seven times the validation rate (PART IX). A production model would need its calibration reviewed regularly. Capacity-based review is less affected by this problem.

I. Time-of-Day Pattern

In E6, time of day was the most important feature by SHAP. The time pattern reflects how PaySim distributes fraud across the day and may not hold for Agribank customers. It is also linked to the time-of-day false alerts reviewed in PART XI.

J. Data Required for a Production FMS

A production FMS would require considerably more data than PaySim provides, including:

- Customer transaction history and risk profile.
- Device information, IP address and geolocation.
- Login and authentication events.
- Changes to mobile number, email address or enrolled beneficiaries.
- Transaction velocity across channels.
- Previous fraud cases, known compromised accounts and blacklists.
- Channel and payment rail information.

These correspond to the fraud management mechanisms identified by BSP: transaction velocity, account and device changes, geolocation, blacklist screening and detection of unusual behavior.

PART XIII: PROPOSED ARCHITECTURE AND BUSINESS APPLICATION

A. Current and Proposed State

The table separates what exists today from what is proposed. It covers only the items relevant to this project.

Area	Current State	Proposed State
Agribank Online	In operation; fund transfers not yet available	Real-time and batch transfers via InstaPay and PESONet (planned for 2027)
Fraud management	Not assessed in this document	An FMS in line with BSP requirements (future initiative)
Machine learning scoring	None	Risk scoring component, demonstrated by this prototype
Data used	Not applicable	PaySim for the prototype; governed Agribank data for any future version

Note: The Bank's existing fraud controls are outside the scope of this capstone. They would need to be documented before the proposed component could be designed for integration.

B. Proposed Risk Scoring Flow

Stage	Proposed Step	Coverage in the Prototype
1	Customer initiates a transfer in Agribank Online	Not covered
2	Transaction details are collected	Simulated using PaySim
3	Business rules are applied (scope filter)	Covered
4	Features are calculated	Covered
5	The model scores the transaction	Covered
6	The score is calibrated	Covered
7	A risk score and level are assigned	Covered
8	Reasons for the score are attached	Covered
9	The alert is routed for manual review	Simulated
10	Review outcomes are used for retraining	Future work

C. Use in Manual Review

The system ranks transactions for review. It does not declare any transaction fraudulent. A sample alert list is shown below.

Transaction	Amount (simulated units)	Risk Score	Priority	Main Reasons
TXN-001	5,200	08	Low	Usual amount for the transaction type
TXN-002	85,000	91	High	Sender account emptied; amount well above usual; new receiving account
TXN-003	12,500	67	Medium	Unusual hour; large amount relative to balance
TXN-004	3,200	12	Low	Usual amount; receiving account used before

Note: The values above are examples and are not model results.

D. Path to a Full FMS

A full FMS would combine this component with rule-based detection, behavioral analytics, device intelligence, velocity controls, blacklist screening and case management. Real-time intervention could be added later, subject to approval.

PART XIV: BUSINESS VALUE AND CONCLUSION

A. Business Value

If later validated on actual bank data, the approach could provide the following benefits:

- **Better fraud detection.** Machine learning may find patterns that fixed rules miss. On PaySim, the realistic model scored about 45 times the rule-based benchmark on PR-AUC.
- **Better use of review time.** Risk scores direct fraud personnel to the highest-risk transactions first.
- **Capacity for higher volumes.** Automated scoring handles large volumes more consistently than manual review alone.
- **Improvement over time.** As confirmed fraud cases accumulate, the model can be retrained on newer patterns.
- **Basis for a future FMS.** The prototype provides a first analytical component that can later be integrated.

B. Conclusion

Fund transfers in Agribank Online will require the Bank to identify and investigate suspicious transactions while they are taking place. This project proposes a fraud detection and risk scoring component as a first step toward that capability.

Using PaySim, the project covers the full machine learning process. It gives particular attention to the low share of fraud, leakage, data availability and the reliability of risk scores. Stress test results are reported alongside headline results, so that the conclusions reflect what machine learning could realistically contribute.

The main finding is the gap between the two views. With every PaySim feature, the models are near perfect, because the simulator leaves clear traces. With only the information Agribank would have when a transfer is submitted, the best model reached a PR-AUC of 0.578. That second figure is the more honest starting point for any discussion of real use.

The prototype is a proof of concept, not a production system. It shows how risk scores could support fraud personnel while decisions remain with them.

C. Recommended Next Steps

Beyond the capstone, the following steps would be needed before any production use:

- Obtain approved access to anonymized Agribank transaction and fraud case data.
- Repeat the method on that data and compare the results with the PaySim findings.
- Run the model in parallel with live processing, without affecting transactions, to observe its performance.
- Establish model governance, validation and monitoring.
- Integrate device, login and channel data, as well as case management.
- Secure internal and regulatory approvals.

ANNEX A: MACHINE LEARNING WORKFLOW

No.	Stage	Main Activities	Output
1	Data collection	Download PaySim; record source and version	Raw dataset
2	Data validation	Data types, missing values, duplicates, extreme values, leakage	Data quality report
3	Exploration	Fraud by type, amount, balance and time; verify known characteristics	Exploration notebook and findings
4	Feature engineering	Balance, amount, time, type and receiving account history features	Feature Sets A to D
5	Feature selection	Mutual information, model-based and permutation importance	Final feature list
6	Dimensionality reduction	PCA on training data	PCA findings
7	Model development	Benchmark, Logistic Regression, Decision Tree, Random Forest, XGBoost; tuning on Set A and Set D	Trained models
8	Evaluation	Metrics on the time-based test set; tests E1 to E6	Comparison table and charts
9	Explainability	SHAP analysis	Charts and alert reasons
10	Bias review	Error rates by transaction group; mitigation test without time features	Review table and discussion
11	Risk scoring	Calibration chosen on validation, 0 to 100 score, capacity rule and threshold comparison	Risk scoring approach
12	Prototype	Transaction, score, alert, manual review	Demonstration

ANNEX B: GLOSSARY OF TERMS

Term	Meaning
AFASA	Anti-Financial Account Scamming Act (Republic Act No. 12010)
Brier score	A measure of how close predicted probabilities are to actual outcomes
Calibration	Adjusting model scores so that predicted probabilities match observed fraud rates
Detection rate	Share of actual fraud that the model flags; also called recall or true positive rate
False positive rate (FPR)	Share of legitimate transactions wrongly flagged
Feature	A field used by the model to make a prediction
FMS	Fraud Management System
Interbank transfer	A transfer to an account at another bank, such as through InstaPay or PESONet
Leakage	Use of information during training that would not be available in actual use
NRPS	National Retail Payment System
On-us transfer	A transfer between two accounts at the same bank
PR-AUC	Area under the precision-recall curve; summarizes how well a model catches fraud while limiting false alerts
Precision@k	Share of actual fraud among the k highest-scoring transactions
SHAP	SHapley Additive exPlanations; shows how much each feature contributed to a prediction
SMOTE	Synthetic Minority Over-sampling Technique; creates synthetic fraud examples to rebalance training data
Stress test	Repeating the evaluation with certain features removed to see how much the model depends on them

ANNEX C: CAPSTONE REQUIREMENTS TRACEABILITY

Capstone Requirement	Where Addressed
Problem definition and framing	PART III
Success metrics	PART IV
Data collection and understanding	PART V
Data preprocessing	PART VI, Section A
Exploratory data analysis	PART VI, Section B
Feature engineering	PART VI, Sections C and D
Feature selection	PART VII, Section A
Dimensionality reduction	PART VII, Section B
Model development and comparison	PART VIII
Explainability	PART X
Bias and fairness assessment	PART XI
Model comparison on realistic features	PART VIII, Section G
Mitigation test	PART XI, Section G
Reproducibility (saved models, configuration, versions)	ANNEX E; GitHub repository
Communication of results	PARTs XIII and XIV; GitHub repository

ANNEX D: REFERENCES

Bangko Sentral ng Pilipinas. (2024). Memorandum No. M-2024-029: Reiterating the Guidelines on Risk Management Systems and Controls to Protect Financial Accounts in relation to Section 6 of Republic Act No. 12010 or the Anti-Financial Account Scamming Act (AFASA).

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ANNEX E: REPRODUCIBILITY

The results in this report were produced by a single script, src/fraud_capstone_pipeline.py, run from start to finish. The script saves the figures quoted in this report to reports/tables/report_numbers.json and the charts to reports/figures/. These outputs were used to fill in the tables and figures in this report, which reduces the risk of copying errors.

Item	Detail
Random seed	42 (sampling, model training and bootstrap)
Data split	Time-based: steps 1 to 322 (training), 323 to 376 (validation), 377 to 743 (test)
Model selection	Validation PR-AUC only. Models within 0.002 of the best are treated as tied, and ties are broken by ease of explanation.
Calibration and thresholds	Chosen on the validation set only (5-fold for calibration). The test set is used once, for reporting.
Saved models	models/*.joblib, including the four Set D models and the calibrators
Configuration	models/run_config.json (settings, feature sets, selected models, thresholds)
Software versions	pandas==2.3.3; numpy==2.2.6; scikit-learn==1.7.2; xgboost==3.2.0; shap==0.49.1; matplotlib==3.10.8
Claim checks	reports/tables/17_report_claim_checks.csv lists each claim made in words in this report and whether the run supports it

To reproduce the results, install the packages listed in requirements.txt, place the PaySim CSV file in data/raw/, and run the script.